Supplemental Materials

Top-Down Identification and Physics-Informed Learning of RC Frames with Dry-Stacked ICEB Infills

This document supplements the paper and contains the derivations, numerical settings and additional results moved from the main text. Equations, tables and figures are numbered with the prefix S; numbers without the prefix refer to the paper.

S1. Material Properties

Table S1 lists the material properties reported by Khan et al. (2026). The assemblage strength f_p of each panel was measured on wallettes of the same construction.

Table S1. Material properties reported by Khan et al. (2026).

Material	Property	Value	COV (%)
Clay brick	Compressive strength (MPa)	11.07	10.2
ICEB unit	Compressive strength (MPa)	3.74	10.6
Mortar	Compressive strength (MPa)	5.65	10.9
Grout	Compressive strength (MPa)	12.34	5.9
Frame concrete	Compressive strength f'_c (MPa)	15.0	5.1
Longitudinal bars	Yield / ultimate strength (MPa)	432 / 617	–
Ties	Yield / ultimate strength (MPa)	415 / 592	–
Expanded metal mesh	Yield / ultimate strength (MPa)	337 / 400	4.2
W1 wallette	Assemblage strength f_p (MPa)	3.00	9.85
W2 wallette	Assemblage strength f_p (MPa)	1.31	8.23
W3 wallette	Assemblage strength f_p (MPa)	1.45	6.10

S2. Frame Model

Each member was represented by a single force-based beam–column element with five Gauss–Lobatto integration points (Spacone et al. 1996; Neuenhofer and Filippou 1997). In contrast to the displacement-based formulation, the force-based element interpolates the section forces $\mathbf{s}(x) = [N(x), M(x)]^{\mathrm{T}}$ from the basic element forces $\mathbf{Q} = \left[N, M_i, M_j\right]^{\mathrm{T}}$,

$$\mathbf{s}(x) = \mathbf{b}(x)\mathbf{Q}, \qquad \mathbf{b}(x) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{x}{l} - 1 & \frac{x}{l} \end{bmatrix} \tag{S1}$$

17 so that equilibrium is satisfied exactly along the element, and the element flexibility and its
18 inverse, the element stiffness, follow by integration of the section flexibility $\mathbf{f}_s = \mathbf{k}_s^{-1}$:

19 $$\mathbf{F} = \int_0^l \mathbf{b}^{\mathrm{T}}(x)\mathbf{f}_s(x)\mathbf{b}(x)\mathrm{d}x \approx \sum_{j=1}^{5} w_j\ \mathbf{b}^{\mathrm{T}}(x_j)\mathbf{f}_s(x_j)\mathbf{b}(x_j), \qquad \mathbf{K} = \mathbf{F}^{-1} \tag{S2}$$

20 The section tangent $\mathbf{k}_s$ is obtained by summation over the fibers as

21 $$\mathbf{k}_s = \sum_i E_{t,i} A_i \begin{bmatrix} 1 & -y_i \\ -y_i & y_i^2 \end{bmatrix}, \qquad \varepsilon_i = \varepsilon_0 - y_i \kappa \tag{S3}$$

22 The concrete followed the Kent–Park–Scott law (Kent and Park 1971; Scott et al. 1982).
23 With compressive stress and strain positive,

24 $$\sigma_c = \begin{cases} Kf'_c \left[2\frac{\varepsilon}{\varepsilon_{c0}} - \left(\frac{\varepsilon}{\varepsilon_{c0}}\right)^2 \right], & \varepsilon \le \varepsilon_{c0} \\ \max\{Kf'_c[1 - Z(\varepsilon - \varepsilon_{c0})],\ \sigma_{cu}\}, & \varepsilon > \varepsilon_{c0} \end{cases} \tag{S4}$$

25 with $\varepsilon_{c0} = 0.002K$, the softening slope Z defined by the residual stress σ_{cu} at the ultimate
26 strain, and linear tension softening beyond $f_t = 0.33\sqrt{f'_c}$. The cover was unconfined ($K = 1$,
27 $\sigma_{cu} = 0.05f'_c$ at $3\varepsilon_{c0}$), whereas the core was assigned $K = 1.05$ and $\sigma_{cu} = 0.2Kf'_c$ at $5\varepsilon_{c0}$; the
28 #3 ties at 150 mm, about half the section depth, provide only nominal confinement. The
29 reinforcement followed the Giuffrè–Menegotto–Pinto law (Menegotto and Pinto 1973; Filippou
30 et al. 1983),

31 $$\sigma^* = b\,\varepsilon^* + \frac{(1-b)\varepsilon^*}{(1+|\varepsilon^*|^R)^{1/R}} \tag{S5}$$

32 with $f_y = 432$ MPa, $E_s = 200$ GPa, a hardening ratio $b = 0.005$ and $R_0 = 18.5$, $c_{R1} =$
33 0.925, $c_{R2} = 0.15$.

34 The moment capacities used in the sway mechanism of the paper (Eq. (3)) follow from strain
35 compatibility. For a rectangular section of width b and depth h with layers of n_k bars at depths
36 d_k, the neutral-axis depth c satisfies axial equilibrium with an equivalent stress block of depth
37 $\beta_1 c$ (ACI Committee 318 2019),

$$0.85f'_c\beta_1 cb + \sum_k n_k A_b \sigma_k = N, \qquad \sigma_k = \mathrm{clip}\left(E_s\varepsilon_{cu}\frac{c-d_k}{c},\ -f_y,\ f_y\right) \tag{S6}$$

with $\varepsilon_{cu} = 0.003$ and $\beta_1 = 0.85$, and the nominal moment about the section centroid is

$$M_n = 0.85f'_c\beta_1 cb\left(\frac{h}{2} - \frac{\beta_1 c}{2}\right) + \sum_k n_k A_b \sigma_k \left(\frac{h}{2} - d_k\right) \tag{S7}$$

S3. Panel Envelope and Strut Implementation

The envelope of the panel in lateral terms passes through four points,

$$\begin{aligned} (V_1, d_1) &= \left(r_{cr}V_p,\ r_{cr}V_p/K_p\right), & (V_2, d_2) &= \left(V_p,\ d_{pk}\right), \\ (V_3, d_3) &= \left([r_{cr} + 0.75(1 - r_{cr})]V_p,\ a_3 d_{pk}\right), & (V_4, d_4) &= \left(r_u V_p,\ a_4 d_{pk}\right) \end{aligned} \tag{S8}$$

Each strut was a truss element between opposite joints of the frame, of length $L_d = \sqrt{L^2 + H^2}$ and inclination $\theta_f = \arctan(H/L)$, with a Pinching4 material (Lowes et al. 2004). The lateral envelope was mapped onto the strut axis by $N = V/\cos\theta_f$ and $\delta = \Delta\cos\theta_f$. A truss element drives its material with the axial strain rather than the elongation, so the deformations were divided by the strut length,

$$\varepsilon_k = \frac{\Delta_k\cos\theta_f}{L_d}, \qquad k = 1, \ldots, 4 \tag{S9}$$

and the tension branch was assigned a negligible strength so that each strut is active only while its diagonal shortens. The hysteretic rules and damage indices of the Pinching4 law are those of Lowes et al. (Lowes et al. 2004); with the exponents fixed at unity, the damage indices read

$$\delta_j = g_{j1}\tilde{d}_{\max} + \frac{g_{j1}}{2}\frac{E_h}{g_E E_{\mathrm{mono}}} \le g_{j,\mathrm{lim}}, \qquad j \in \{K, D, F\} \tag{S10}$$

in which $\tilde{d}_{\max}$ is the normalized maximum deformation, E_h the hysteretic energy dissipated so far and E_{mono} the area under the envelope.

Struts. Truss elements of unit area between opposite joints, so that the material carries force rather than stress; Pinching4 with an envelope in compression only and a tension branch of

negligible strength; damage exponents fixed at unity and limits $g_{K,\mathrm{lim}} = 0.90$, $g_{D,\mathrm{lim}} = 0.80$, $g_{F,\mathrm{lim}} = 0.85$.

Solution. Displacement control at the beam–column joint with a Krylov–Newton algorithm and a displacement-increment test of 10^{-6} mm; on failure the step was subdivided successively into 10, 50 and 200 sub-steps, with the tolerance relaxed to 10^{-4}, 10^{-3} and 10^{-2} and the algorithm changed to Newton with bisection line search and finally modified Newton. Pushover analyses to a drift of 5% (at most 150 mm) in 220–250 steps; cyclic analyses with two cycles at each amplitude in increments of 1.5 mm. Each analysis ran in its own process with a time limit, and a crashed or non-converged analysis was recorded as rejected. The analyses were run with OpenSeesPy (Zhu et al. 2018), and the learners were implemented with scikit-learn (Pedregosa et al. 2011), XGBoost (Chen and Guestrin 2016) and PyTorch (Paszke et al. 2019).

S4. Panel Modulus and Cyclic Calibration

The initial stiffness of each panel is the measured stiffness of its specimen less that of the frame at the same drift. Because the strut width depends on the modulus through λ_h, the inversion of the panel stiffness relation is implicit,

$$E_{inf}\, w\big(E_{inf}\big) = \frac{K_p\, d_{inf}}{t_{inf} \cos^2\theta} \tag{S11}$$

and since the left-hand side increases monotonically as $E_{inf}^{0.9}$, it has a unique root, which was found by bracketing. Table **S**2 lists the results. The modulus ratios $n_E = E_{inf}/f_p$ differ between the panels by a factor of more than six, and the differences are physically ordered. The dry-stacked panel has the lowest ratio, 200, because its joints must close before the units bear on one another. The mortar-bonded panel has 447, below the short-term value of 1000 adopted in EN 1996-1-1 (CEN 2005); the difference may be attributed to the grouted gaps and the joints next to

the frame, which add flexibility that a wallette does not have. The meshed panel has 1305, because its two 25 mm plaster layers act as stiff continuous skins on the dry core and the mesh restrains the opening of the joints.

Table S2. Panel stiffness and modulus identified from the initial stiffness of the specimens.

Specimen	$K_{0,\exp}$ (kN/mm)	K_p (kN/mm)	$K_p/K_{0,\exp}$ (%)	E_{inf} (MPa)	n_E	w (mm)
W1	26.83	14.08	52	1342	447	302
W2	15.98	3.23	20	262	200	355
W3	35.29	22.54	64	1892	1305	286

With the envelope fixed, the hysteretic parameters of the strut material were fitted to the normalized secant stiffness and the cumulative dissipated energy of each specimen. The frame and the envelope were held at their identified values, and the seven parameters $[r_\Delta, r_F, u_F, g_{K1}, g_{D1}, g_{F1}, g_E]$ of the Pinching4 law were fitted separately for each panel, because the three masonry systems dissipate energy by different mechanisms: cracking and crushing of the mortar joints, frictional slip on the dry joints, and bridging of cracks by the mesh. The objective was

$$J_c = \frac{1}{0.01\, n_a} \sum_{a=1}^{n_a} \left[\frac{K_a}{K_1} - \left(\frac{K_a}{K_1} \right)^{\exp} \right]^2 + 2 \left(\frac{E_{cum}}{E_{cum}^{\exp}} - 1 \right)^2 \tag{S12}$$

where K_a is the secant stiffness at the first excursion to amplitude a and K_1 that at the first amplitude. Table **S**3 lists the fitted parameters and Fig. **S**1 the computed and measured responses.

Table S3. Hysteretic parameters of the strut material and the fit to the cyclic response.

Specimen	r_Δ	r_F	u_F	g_{K1}	g_{D1}	g_{F1}	g_E	RMSE of K/K_1	E_{cum} (kJ)	$E_{cum}^{\exp}$ (kJ)
W1	0.37	0.18	0.05	0.31	0.16	0.20	9.8	0.012	156.4	118.3
W2	0.46	0.10	0.05	0.31	0.17	0.20	9.2	0.057	168.7	96.1
W3	0.36	0.19	0.05	0.31	0.15	0.21	9.9	0.024	121.0	145.9

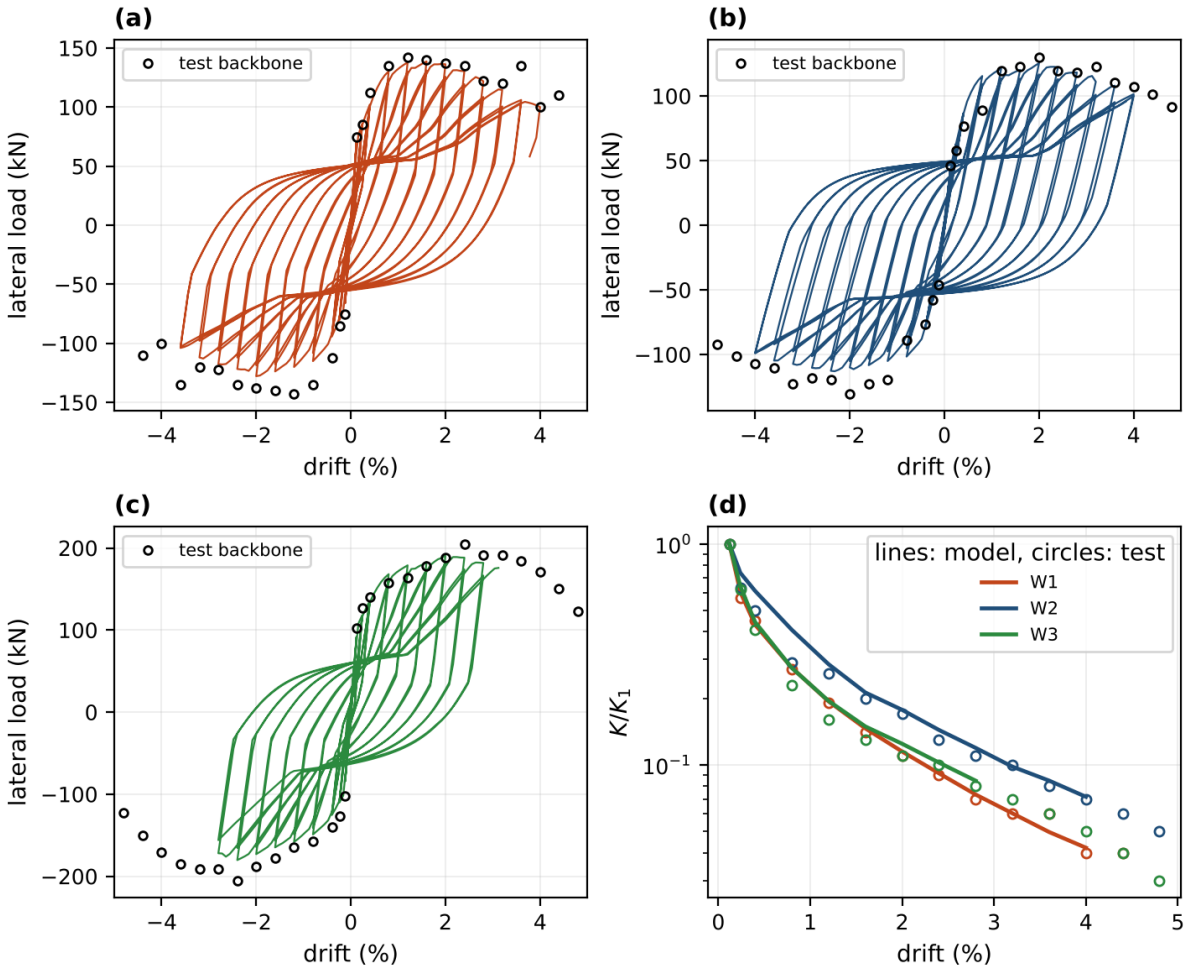


Fig. S1. Cyclic response computed with the identified envelope and the fitted hysteretic parameters: (a)–(c) hysteresis loops of the three systems; (d) normalized secant stiffness against drift.

The normalized stiffness degradation is reproduced closely, with root-mean-square deviations of 0.012, 0.057 and 0.024 in K_a/K_1 (Fig. S1(d)). The dissipated energy is not. The cyclic analyses terminated by loss of convergence before the last amplitudes of the tests, at 100 mm for the two unreinforced systems and at 78 mm for the meshed one. Even so, the analyses of the mortar-bonded and dry-stacked systems dissipated 32% and 76% more energy than was measured over the complete tests; the energy of the meshed system, 121 kJ up to 78 mm, cannot be compared with the measured total. The excess does not originate in the struts. Under the same

protocols the bare frame alone dissipates 181 and 221 kJ, more than the measured totals of the mortar-bonded and dry-stacked specimens, 118 and 96 kJ. A fiber model with perfect bond between bars and concrete produces full hysteresis loops at the hinges, whereas the pinching caused by bar slip and shear cracking in a lightly detailed frame removes a large part of that area (Zhao and Sritharan 2007). Since the frame model was fixed before the identification, the energy was not used to adjust it, and the conclusions of this study rest on the envelope, which the fiber model reproduces, and not on the dissipated energy. A frame model with bond-slip or pinching hinges would be required for a cyclic or dynamic extension.

S5. Learners and Settings

A surrogate of the system model maps the eleven variables of Table 5 to the peak lateral load. Mechanical knowledge of the system can be placed in it at five levels, and the learners compared here were chosen so that each level is represented (Table **S**4). At the level of the data, quantities computed by the mechanics are supplied as inputs. At the level of the target, the learner is asked to predict the part of the response that the mechanics cannot supply, here the compatibility factor of Eq. (10). At the level of the prior, the mechanics provides the mean about which a probabilistic learner expresses its uncertainty. At the level of the constraints, the signs of the derivatives that the mechanics prescribes are imposed or penalised. At the level of the structure, the form of the model is the form of the mechanics, a combination of the frame and the panel. The terms used for these constructions in the literature vary (Karpatne et al. 2017; Willard et al. 2022; Karniadakis et al. 2021); the classification by level is adopted here because it states what each construction assumes.

Table S4. Learners compared, grouped by the level at which mechanics enters.

Level	Learner	Inputs	Output	What the mechanics supplies

Level	Learner	Inputs	Output	What the mechanics supplies
None	OLS-2, SVR, GPR, RF, GBT, MLP	11 variables	V_{pk}	–
Data	PI-F	11 variables + 17 mechanical quantities	V_{pk}	V_{fr}, V_p, V_c, V_m, σ_s, σ_{sl}, mode, w, $\lambda_h H$, K_p, d_{pk}, …
Target	PI-R	as PI-F	κ	the scale $V_\Sigma = V_{fr} + V_p$
Prior	PI-G	11 variables + 6 groups	$\ln\kappa$	the prior mean V_Σ and its uncertainty
Constraint	PI-C, PI-L	11 variables	V_{pk}	the signs of $\partial V_{pk}/\,\partial x_j$
Structure	PI-A, PI-N	11 variables + 6 groups	V_{pk}	the decomposition into frame and panel

Physical admissibility was measured by the violation rate

$$\nu = \frac{100}{n_p|\mathcal{S}|}\sum_{j\in\mathcal{S}}\sum_{i=1}^{n_p}\mathbb{1}\left[s_j\left(\hat{f}\left(\mathbf{x}_i + h_j\mathbf{e}_j\right) - \hat{f}(\mathbf{x}_i)\right) < -\tau\left|\hat{f}(\mathbf{x}_i)\right|\right] \tag{S13}$$

evaluated on $n_p = 600$ samples with steps $h_j = 0.06\,\mathrm{sd}(x_j)$ and a tolerance $\tau = 2\times10^{-3}$, below which a change is treated as numerical noise.

Learners. OLS-2: ridge regression on the standardized quadratic expansion, α selected from 40 values in $[10^{-4}, 10^{3}]$ by cross-validation. SVR: radial basis kernel, $C = 30$, $\epsilon = 0.01$ on the standardized target. GPR: constant times anisotropic RBF plus white noise, length scales in $[10^{-2}, 10^{3}]$, two optimizer restarts. RF: 500 trees, minimum leaf size 2. GBT, PI-F, PI-R, PI-C: XGBoost (Chen and Guestrin 2016) with 900 trees, learning rate 0.045, maximum depth 5, row and column subsampling 0.85, minimum child weight 2, L_2 regularization 1.2. MLP: three hidden layers of 96, 96 and 48 ReLU units, L_2 penalty 10^{-4}, early stopping. PI-G: as GPR on $\ln\kappa$ with 17 inputs and one optimizer restart. PI-L, PI-A, PI-N: networks of two hidden layers with 64 SiLU units, one for PI-L and two for PI-A and PI-N, Adam with initial learning rate 4×10^{-3}, weight decay 10^{-5}, batch size 256, 2600 epochs with cosine annealing, gradient clipping at 5.

S6. Signs of the Derivatives of the System Capacity

The sign constraints used for the learners in the paper follow from the panel and frame models. Substituting Eq. (5) into Eq. (6) and multiplying numerator and denominator by $1 + 1.5\mu_j\lambda_u$ gives the sliding limit in terms of the joint properties directly,

$$\sigma_{sl} = \frac{c_j+\mu_j\sigma_v}{\sin\theta\cos\theta(1+1.5\mu_j\lambda_u)-\mu_j\sin^2\theta} = \frac{c_j+\mu_j\sigma_v}{\sin\theta\cos\theta-\mu_j\sin\theta\cos\theta(\tan\theta-1.5\lambda_u)} \quad \text{(S14)}$$

so that, writing D for the denominator,

$$\frac{\partial\sigma_{sl}}{\partial c_j} = \frac{1}{D} > 0, \qquad \frac{\partial\sigma_{sl}}{\partial\sigma_v} = \frac{\mu_j}{D} > 0, \qquad \frac{\partial\sigma_{sl}}{\partial\mu_j} = \frac{\sigma_v D+(c_j+\mu_j\sigma_v)\sin\theta\cos\theta(\tan\theta-1.5\lambda_u)}{D^2} \quad \text{(S15)}$$

The first two derivatives are positive wherever sliding is possible ($D > 0$). The third is positive provided $\tan\theta > 1.5\lambda_u$, which over the design space holds with a wide margin, since $\tan\theta \geq 0.6$ and $1.5\lambda_u = 0.5$. The condition is the same as that which governs the effect of joint friction on the shear resistance of a masonry wall, and for the same reason: the stepped crack makes the friction coefficient reduce the effective cohesion while it increases the frictional resistance.

Because $\sigma_s = \min(f_p, \sigma_{sl})$, the strut stress is non-decreasing in c_j, μ_j and σ_v. The strut width varies with the thickness as $t_{inf}^{-0.1}$ through λ_h, so $V_c \propto t_{inf}^{0.9}$ increases with thickness; the mesh term increases with q_m. The capacity of the frame, Eq. (3), increases with f'_c, f_y and the area of the column bars, and with the axial load for axial forces below the balanced point, which covers the whole design space. A higher f'_c also stiffens the columns, which lowers λ_h and widens the strut. Each of these variables therefore increases at least one component. The pre-compression is the one case in which the two components respond in opposite directions: it raises the sliding resistance of the panel but also the axial load ratio of the columns, which lowers the resistance they retain after their peak (Eq. (14)). The sweeps of the paper show that the first effect

dominates until crushing takes over, after which the capacity is practically constant, so the constraints $s_j = +1$ are imposed on all eight.

The remaining three variables act in opposite directions. With n_E held fixed, the assemblage strength enters both the strut stress and the modulus, $E_{inf} = n_E f_p$. Where crushing governs, $V_c \propto f_p\, w \propto f_p^{0.9}$ increases; where sliding governs, $\sigma_s = \sigma_{sl}$ does not depend on f_p and $V_c \propto w \propto f_p^{-0.1}$ decreases. The modulus ratio scales the strut stiffness as $E_{inf}^{0.9}$ but its width as $E_{inf}^{-0.1}$, so that at a fixed strength a stiffer panel is slightly weaker, while a stiffer system attracts load at smaller drifts. The aspect ratio increases θ, which raises σ_{sl} through $\tan\theta$ in Eq. (S14), and increases the column length H, which lowers V_{fr} in proportion to $1/H$. No sign is imposed on any of the three.

S7. Equations from the Shapley Attribution

Two closed forms were built from the attribution. In Equation A each main effect of Fig. 6 is represented by the best of six shapes, u, $\sqrt{u}$, $\ln(1+3u)$, u^2, $u^{1.5}$ and $1/(1+3u)$, where $u_j = \left(x_j - x_{j,\min}\right)/\left(x_{j,\max} - x_{j,\min}\right)$ maps each variable onto its sampled range, and the coefficients are refitted jointly by least squares:

$$V_A = \beta_0 + \sum_{j=1}^{11} \beta_j \; h_j\left(u_j\right) \tag{S16}$$

Equation A is additive and cannot represent the products that the interactions require. Equation B adds the mechanism groups of the two components, each of which is an exact product of variables,

$$V_B = V_A + \gamma_{fr} V_{fr} + \gamma_c \, \sigma_s w t_{inf} \cos\theta + \gamma_m q_m L_{inf} + \gamma_{ax} \sigma_v L_{inf} t_{inf} \tag{S17}$$

with the coefficients obtained by ridge regression. Equation A reaches $R^2 = 0.814$ and Equation B $R^2 = 0.996$ on the database.

S8. Database and One-Factor Sensitivity

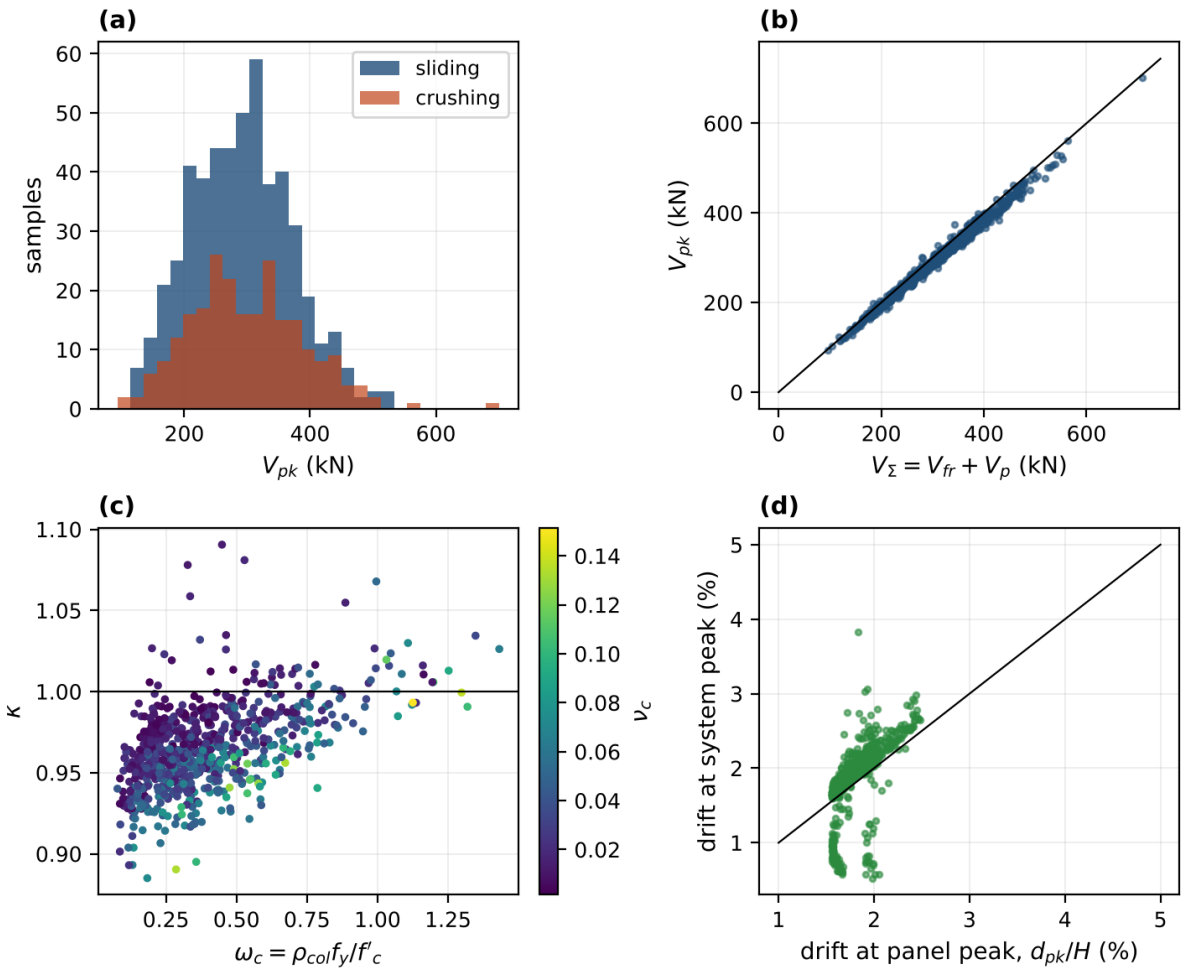


Fig. S2. The parametric database: (a) peak lateral load by governing mode of the panel; (b) peak load against the sum of the frame and panel capacities; (c) compatibility factor κ against the mechanical reinforcement ratio of the columns, colored by their axial load ratio; (d) drift at the panel peak against the drift at the peak of the system.

Fig. **S**3 shows the response of the system model to each variable about the EMM-reinforced specimen, with all other variables held at their tested values. The sweeps confirm the signs used for the learners in the paper. The mesh yield force has the largest effect, raising the capacity by 80% of its mean over its range, followed by the column reinforcement ratio (37%), the pre-

compression (31%) and the panel thickness (25%). The pre-compression and the cohesion both act through the sliding stress, and their effect saturates, at $\sigma_v \approx 0.29$ MPa and $c_j \approx 0.09$ MPa respectively, when σ_{sl} reaches f_p and crushing takes over. The assemblage strength changes the capacity by only 4%, and not monotonically, for the reason given in the paper, and the response to the panel aspect ratio is non-monotonic: as the panel becomes taller, the frame capacity falls in proportion to $1/H$ while the sliding resistance of the steepening strut rises, until at $H_{inf}/L_{inf} \approx 1.5$ the strut stress reaches f_p, crushing takes over and the panel resistance stops increasing.

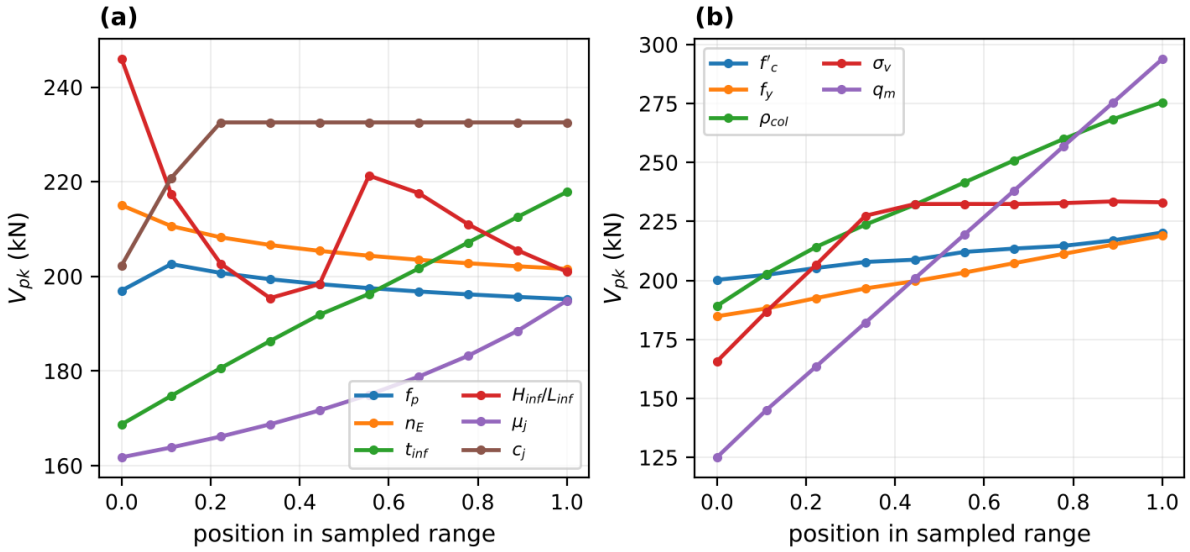


Fig. S3. One-factor sensitivity of the peak lateral load about the EMM-reinforced specimen; each panel groups variables of one component: (a) masonry and joints; (b) frame, mesh and loading.

S9. Notation

Table S5. Principal symbols.

Symbol	Meaning
a_3, a_4	displacement ratios of the softening and residual points of the panel envelope
C_d, C_m	seating term of dry joints; mesh term of the displacement at peak
c_j, c_m	cohesion of the bed joint; reduced cohesion on the stepped crack
d_{inf}	length of the panel diagonal
d_{pk}	lateral displacement of the panel at its peak

Symbol	Meaning
E_c, E_{inf}	elastic modulus of the frame concrete and of the panel
f'_c, f_y	compressive strength of the frame concrete; yield strength of its reinforcement
f_p	compressive strength of the masonry assemblage
H, L	centerline height and span of the frame
H_{inf}, L_{inf}, t_{inf}	height, length and thickness of the panel
I_c	effective moment of inertia of the column
K_{fr}, K_p	initial lateral stiffness of the frame and of the panel
k_m	mobilized fraction of the mesh yield force
M_c, M_b	nominal moment capacities of the column and the beam
n_E	modulus ratio E_{inf}/f_p
q_m	yield force of the mesh per unit length of panel, both faces
r_{cr}, r_u	force ratios of the cracking and residual points of the panel envelope
V_c, V_m, V_p	resistance of the masonry strut; of the mesh; of the panel
V_{fr}	sway capacity of the frame
V_{pk}, V_Σ	peak lateral load of the system; sum $V_{fr} + V_p$
w	width of the equivalent strut
α_p, α_{fr}	fractions of the panel and frame capacities mobilized at the system peak
δ_{fr}, γ, ρ	coefficients of the compatibility equation
ε_p	strut strain at peak
η_c	efficiency of the strut
θ	inclination of the strut
κ	compatibility factor V_{pk}/V_Σ
λ_h	relative stiffness parameter of Stafford Smith and Carter
λ_u	aspect ratio of the masonry unit
μ_j, μ_m	friction coefficient of the bed joint; reduced coefficient on the stepped crack
ν	violation rate of the sign constraints
ρ_{col}	longitudinal reinforcement ratio of the columns
σ_s, σ_{sl}	limiting stress of the strut; stress at which the bed joints slide
σ_v	vertical pre-compression of the panel
ψ_j	reduction factor of the stepped crack
ω_m	mechanical mesh ratio $q_m/(t_{inf}f_p)$

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